

Newman-Penrose Formalism Calculations

Stephani et al., Equation (31.50)

Metric

$$ds^2 = -Kv^2 du \otimes du - du \otimes dv - dv \otimes du + \left(\frac{16}{K^2 x^4 + K^2 y^4 + 8 K x^2 + 2 (K^2 x^2 + 4 K) y^2 + 16} \right) dx \otimes dx + \left(\frac{16}{K^2 x^4 + K^2 y^4 + 8 K x^2 + 2 (K^2 x^2 + 4 K) y^2 + 16} \right) dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, v, x, y)$$

Null Tetrad

Covariant components

$$l = (1) \, du$$

$$n = \left(\frac{1}{2} K v^2 \right) du + (1) \, dv$$

$$m = \left(\frac{2 \sqrt{2}}{K x^2 + K y^2 + 4} \right) dx + \left(\frac{2i \sqrt{2}}{K x^2 + K y^2 + 4} \right) dy$$

$$\bar{m} = \left(\frac{2 \sqrt{2}}{K x^2 + K y^2 + 4} \right) dx + \left(-\frac{2i \sqrt{2}}{K x^2 + K y^2 + 4} \right) dy$$

Contravariant components

$$l = (-1) \, dv$$

$$n = (-1) \, du + \left(\frac{1}{2} K v^2 \right) \, dv$$

$$m = \left(\frac{1}{8} \sqrt{2} (K x^2 + K y^2 + 4) \right) \, dx + \left(\frac{1}{8} \sqrt{2} (i K x^2 + i K y^2 + 4i) \right) \, dy$$

$$\overline{m} = \left(\frac{1}{8} \sqrt{2} (K x^2 + K y^2 + 4) \right) \, dx + \left(\frac{1}{8} \sqrt{2} (-i K x^2 - i K y^2 - 4i) \right) \, dy$$

Directional Derivatives

$$D = -\partial_v$$

$$\Delta = -\partial_u + \frac{1}{2} K v^2 \partial_v$$

$$\delta = \frac{1}{8} \sqrt{2} (K x^2 + K y^2 + 4) \partial_x + \frac{1}{8} \sqrt{2} (i K x^2 + i K y^2 + 4i) \partial_y$$

$$\bar{\delta} = \frac{1}{8} \sqrt{2} (K x^2 + K y^2 + 4) \partial_x + \frac{1}{8} \sqrt{2} (-i K x^2 - i K y^2 - 4i) \partial_y$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = 0$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{1}{8} \sqrt{2} K (x - i y)$$

$$\beta = -\frac{1}{8} \sqrt{2} K (x + i y)$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = -\frac{1}{2} K v$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{1}{2} K$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = 0$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = 0$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "O"